Drug Repurposing Hypothesis Assessment: NLRP3 Inflammasome Inhibition (Dapansutrile) in Amyotrophic Lateral Sclerosis

Every factual claim is cited to a primary source by PMID with a verbatim quote. "ESTABLISHED" marks findings reported in the cited source; "INFERENCE" marks my own reasoning built on those findings.

Summary

The hypothesis that an NLRP3 inflammasome inhibitor such as **dapansutrile (OLT1177)** would benefit patients with Amyotrophic Lateral Sclerosis (ALS) by blocking microglial NLRP3 → caspase-1 → IL-1β/IL-18 signaling triggered by misfolded SOD1 and TDP-43 is **mechanistically well-grounded but therapeutically unproven**. The target pathway is unambiguously active in ALS: inflammasome machinery (NLRP3, ASC, caspase-1) and its products (IL-1β, IL-18) are elevated in human ALS post-mortem CNS tissue, in the blood and CSF of living patients, and in multiple animal models. Both of the proteinopathies central to ALS — misfolded SOD1 and TDP-43 — have been shown experimentally to activate the microglial NLRP3 axis. In SOD1 mouse models, interrupting the inflammasome output (via caspase-1/IL-1β deficiency, IL-1 receptor antagonism, or NLRP3 pathway down-modulation) extends survival and improves motor function. Dapansutrile itself is an orally bioavailable, NLRP3-specific, human-safe inhibitor that reaches the brain and rescues cognition in an Alzheimer's model, making the drug pharmacologically plausible for CNS use.

However, the therapeutic case has serious unresolved weaknesses. **No NLRP3 inhibitor has ever been trialed in ALS.** The only direct human test of the broader IL-1 axis — a pilot study of the IL-1 receptor antagonist anakinra — showed no slowing of disease progression and was confounded by near-universal anti-drug antibody formation. Broad-spectrum anti-inflammatory and immunosuppressive strategies have a long history of failure in ALS, reflecting a mismatch between non-selective drugs and the stage- and genotype-dependent, sometimes neuroprotective, immune states of the disease. Parallel inflammatory arms (NF-κB,

AP-1, cGAS-STING) that an NLRP3 inhibitor would not block create redundancy that could blunt efficacy. The dominant NLRP3-expressing cell in some human studies was the astrocyte, not the microglia, qualifying the precise cellular logic of the hypothesis.

On balance, the mechanism is active and the drug is reachable, but efficacy in ALS is genuinely uncertain. The single most informative next step is **not** a broad all-comers trial but a **biomarker-stratified** trial enrolling patients with demonstrable inflammasome activation (e.g., elevated CSF IL-18), with a CSF IL-18/IL-1β target-engagement readout to confirm the drug is doing what it is designed to do before judging clinical endpoints. This would test the hypothesis efficiently and avoid repeating the historical pattern of diluting a real effect in an unselected population.

Key Findings

Finding 1 — The NLRP3 inflammasome is demonstrably active in ALS patient CNS tissue

(ESTABLISHED)

Two independent human post-mortem studies, supported by mouse models, establish that inflammasome machinery and its products are elevated in affected ALS tissue. Johann and colleagues found that in SOD1-G93A mice NLRP3, ASC, caspase-1 and IL-1β were detectable pre-symptomatically (from 9 weeks) and rose with disease, and critically that *"In human ALS tissue, we also found increased NLRP3, ASC, IL18 and active caspase 1 levels compared to control patients"* (PMID: 26200799). Van Schoor and colleagues mapped inflammasome and pyroptosis effector proteins in post-mortem brain and spinal cord, reporting that they *"determined the expression and distribution of the inflammasome and pyroptosis effector proteins in post-mortem brain and spinal cord from ALS patients (n = 25) and controls (n = 19)"* and linking increased microglial pyroptosis to neuronal loss in the motor cortex (PMID: 35867112).

Important qualification (ESTABLISHED): In the Johann study, *"Spinal cord astrocytes were identified as the major cell type expressing NLRP3 components"* (PMID: 26200799). This directly complicates the hypothesis's framing of microglia as the dominant NLRP3-expressing compartment — at least in human spinal cord, astrocytes may carry the larger share, which matters for predicting which cells a drug must reach.

Finding 2 — Causal preclinical evidence: blocking the inflammasome output slows disease in SOD1 mice

(ESTABLISHED)

Meissner and colleagues provided the foundational causal evidence that inflammasome output is not merely a bystander. They showed that cytoplasmic mutant SOD1 is *"sensed by an ASC containing inflammasome"* and, decisively, that *"Deficiency in caspase-1 or IL-1beta or treatment with recombinant IL-1 receptor antagonist (IL-1RA) extended the lifespan of G93A-SOD1 transgenic mice and attenuated inflammatory pathology"* (PMID: 20616033). This is the strongest single piece of causal preclinical support for the hypothesis. **(INFERENCE)** Note, however, that it demonstrates engagement of *an ASC-containing* inflammasome rather than proving NLRP3 specificity in that setting. The NLRP3 specificity is supplied by complementary work (Deora 2020) showing that *"Pathogenic protein aggregates such as β-amyloid and α-synuclein trigger microglial NLRP3 activation, leading to caspase-1 activation and IL-1β secretion"* (PMID: 31596526).

Finding 3 — The only human IL-1-axis trial in ALS was negative

(ESTABLISHED)

This is the most important piece of evidence *against* straightforward optimism. A one-year pilot study treated 17 ALS patients with anakinra (IL-1 receptor antagonist) 100 mg daily against a 47-patient historical cohort. The authors reported that *"we could not detect a significant reduction in disease progression during the same period in patients treated with ANA compared to controls as measured by the ALSFRSr"* (PMID: 26444282). Interpretation is further clouded by immunogenicity: *"Sixteen out of the 17 patients (94%) developed antibodies against ANA"* (PMID: 26444282). **(INFERENCE)** This negative result tests a point downstream of NLRP3 (IL-1 receptor signaling) in an unselected population, so it does not definitively refute upstream NLRP3 inhibition — but it is a direct, sobering signal that blocking the IL-1 axis alone did not change the clinical course.

Finding 4 — Dapansutrile reaches the brain, is CNS-efficacious in a neurodegeneration model, and is clinically safe peripherally

(ESTABLISHED)

The drug-level plausibility is strong. Lonnemann and colleagues used *"pharmacological inhibition of the NLRP3 inflammasome using dapansutrile (OLT1177), an oral NLRP3-specific inhibitor that is safe in humans"* and showed that oral dosing rescued learning and memory in

APP/PS1 Alzheimer's mice, noting that *"The Morris water maze test revealed an impaired learning and memory ability of 9-mo-old APP/PS1 mice"* that was reversed by treatment (PMID: 33257576). Clinically, a phase 2a gout trial confirmed oral target engagement and tolerability across a wide dose range, establishing that *"Dapansutrile inhibits the NLRP3 inflammasome and subsequent activation of IL-1β"* with efficacy and safety in an IL-1β-driven disease (PMID: 33005902). **(INFERENCE)** Together these establish that the specific drug named in the hypothesis is orally bioavailable, brain-accessible, NLRP3-selective, and human-safe — removing the most obvious pharmacological objections, although CNS efficacy has only been shown in an Alzheimer's (not ALS) model.

Finding 5 — Neuroinflammation is a tractable ALS target, but no approved therapy hits NLRP3 specifically

(ESTABLISHED)

The masitinib program demonstrates that targeting neuroinflammation can change ALS clinical outcomes. Masitinib *"was effective in phase 2/3 and 3 clinical trials (CTs) by increasing overall survival and delaying motor, bulbar, and respiratory function deterioration"* (PMID: 36774721), and preclinically *"Masitinib-induced mast cell reduction resulted in a 35% decrease in NMJ denervation and reduced motor deficits as compared with vehicle-treated rats"* (PMID: 29046475). **(INFERENCE)** This is encouraging proof-of-concept that immune-targeted therapy is viable in ALS — but it also highlights that the clinically validated axis to date is mast-cell/microglial tyrosine-kinase signaling, *not* NLRP3, which remains without a dedicated ALS therapy.

Finding 6 — Human in-vivo biomarker evidence identifies an active, male-enriched inflammasome subgroup

(ESTABLISHED)

Living-patient data elevate this beyond post-mortem description. In a two-cohort study of newly diagnosed ALS patients, among 12 CSF neuroimmune biomarkers, *"Male ALS patients showed markedly elevated CSF GFAP, IL-6, and IL-18 compared with female ALS patients and HCs"* (PMID: 42425169). Independent peripheral-blood work found that *"NLRP3 mRNA expression was significantly higher in ALS-cultivated PBMCs"* (PMID: 42646494), and a spontaneous canine ALS model (degenerative myelopathy) showed that *"mRNA expression of interleukin (IL)-18 and nod-like receptor protein 3 was significantly increased compared with intervertebral disk herniation controls"* (PMID: 41832208). **(INFERENCE)** CSF IL-18 is a

candidate stratifying biomarker, and the sex enrichment suggests inflammasome activity is concentrated in an identifiable subgroup rather than uniform across all patients — a crucial design consideration for any trial.

Finding 7 — ALS immune states are stage/genotype-dependent, and broad immunosuppression has repeatedly failed

(ESTABLISHED)

A 2026 precision-medicine review frames the central risk: ALS immune cells *"adopt context-dependent states that can be neuroprotective or neurotoxic depending on disease stage and genetic background,"* and *"The repeated failure of broad-spectrum immunosuppressive trials reflects a fundamental mismatch between the non-selective interventions and the mechanistically distinct immune states of diseases"* (PMID: 42638120). **(INFERENCE)** This is the strategic argument for stratification and against an untargeted trial: a drug given at the wrong stage, or to patients whose microglial state is net-protective, could be neutral or even harmful.

Finding 8 — Down-modulating NLRP3/caspase-1/ASC improves survival and motor function in SOD1 mice

(ESTABLISHED)

Two independent SOD1-G93A studies using electroacupuncture to suppress the inflammasome reinforce the causal chain. One reported that *"EX-B2 EA improves limb function and prolongs survival in ALS mice, with its underlying mechanism potentially involving the alleviation of neuroinflammation through inhibition of NLRP3 inflammasome activation"* (PMID: 42621714); a companion study attributed delayed progression to *"modulating NLRP3/Caspase-1/ASC signaling pathway and inhibiting the inflammatory response, thereby attenuating the inflammatory injury of lumbar spinal cord"* (PMID: 42864293). **(INFERENCE)** These are indirect (the intervention is not a pharmacological NLRP3 inhibitor), but they converge with the genetic evidence that reducing NLRP3-axis activity benefits SOD1 mice.

Finding 9 — Misfolded TDP-43 activates the microglial NLRP3 inflammasome, but via a trigger with parallel arms

(ESTABLISHED)

Zhao and colleagues provided direct mechanistic support for the TDP-43 half of the hypothesis. They showed that *"the NLRP3 inflammasome was induced and functional caspase-1 was produced augmenting the release of mature IL-1β"* in microglia exposed to TDP-43 (PMID: 26222336). Critically for the failure analysis, the same study found that *"This response to TDP-43 was mediated by its interaction with the microglial surface CD14 receptor and subsequent stimulation of the NF-κB and AP-1 pathways, as well as the intracellular inflammasome"* (PMID: 26222336). **(INFERENCE)** The NF-κB and AP-1 arms would be untouched by an NLRP3 inhibitor — built-in redundancy that could limit a single-axis drug's effect.

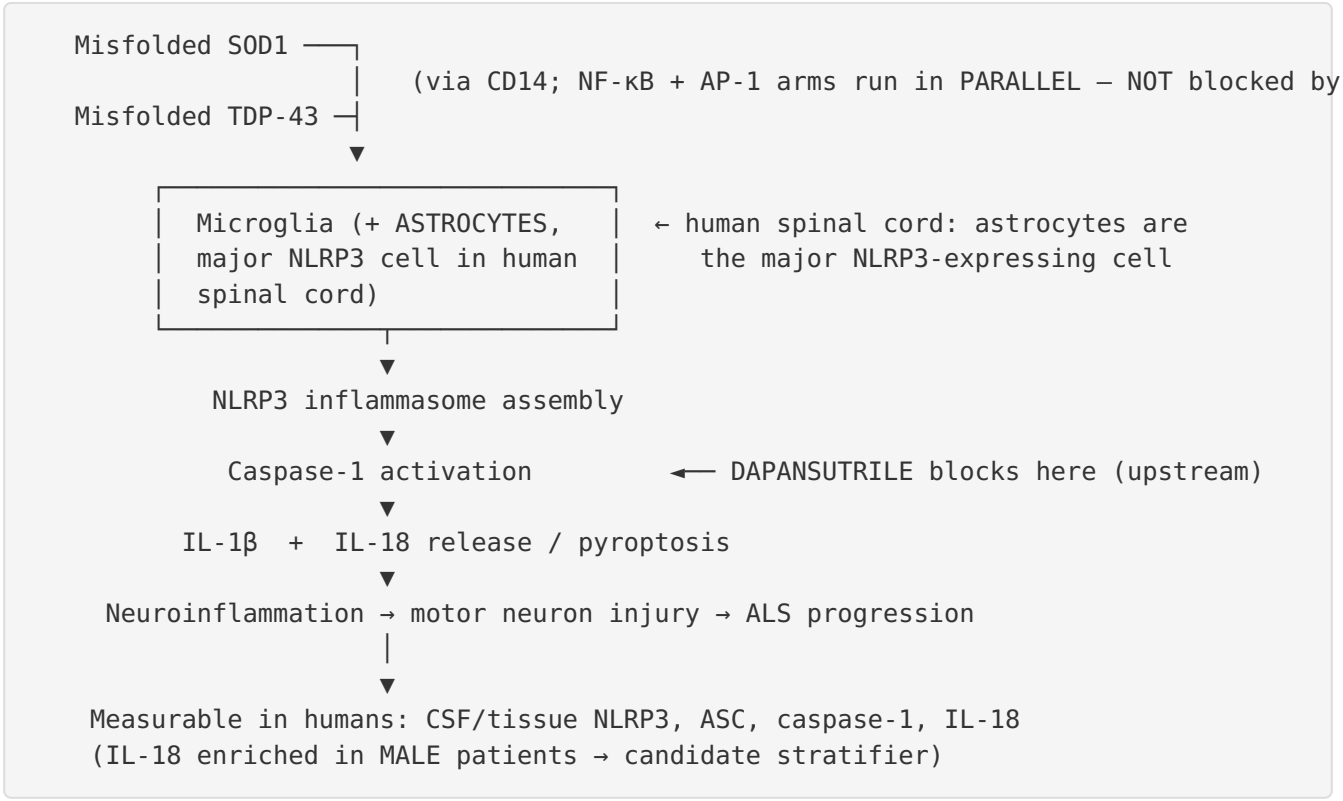
Finding 10 — Safety is acceptable short-term, but infection risk is population-specific

(ESTABLISHED)

NLRP3/IL-1 inhibition has been tolerated: anakinra in ALS caused *"Only mild adverse events occurred in ALS patients treated with ANA"* over one year (PMID: 26444282), and dapansutrile recorded *"No serious adverse events during the study were recorded"* in heart failure (PMID: 33235030) and was tolerated across 100–2000 mg/day in gout (PMID: 33005902). However, the ALS population is uniquely vulnerable to respiratory infection: respiratory insufficiency *"is the commonest cause of death in ALS"* (PMID: 21082322). **(INFERENCE)** Because the NLRP3/IL-1/IL-18 axis is a component of antimicrobial host defense, chronic suppression in a population prone to aspiration pneumonia warrants active infection surveillance, even if short-term tolerability looks benign.

Mechanistic Model / Interpretation

The evidence assembles into a coherent, testable causal chain, with the points of greatest strength and greatest vulnerability both identifiable:



Where the chain is strong: Every node from trigger to output has direct experimental support in ALS-relevant systems. The trigger→NLRP3 link is shown for both SOD1 (P 20616033) and TDP-43 (P 26222336); the output is elevated in human tissue (P 26200799), blood (P 42646494) and CSF (P 42425169); and reducing the output extends survival in SOD1 mice (PMIDs 20616033, 42621714, 42864293).

Dapansutrile acts at caspase-1 activation, upstream of both IL-1β and IL-18, and reaches the brain
(P 33257576).

Where the chain is vulnerable:

1. **Redundancy at the trigger.** TDP-43 drives NF-κB and AP-1 in parallel to the inflammasome (P 26222336). Blocking NLRP3 leaves these intact.
2. **Cell compartment.** If astrocytes — not microglia — are the dominant NLRP3 source in human spinal cord (P 26200799), the hypothesis's microglia-centric logic is only partly correct, though dapansutrile is not cell-type-restricted and would still act on astrocytic NLRP3.
3. **Downstream test was negative.** Blocking IL-1 receptor signaling (anakinra) did not slow ALS (P 26444282). NLRP3 inhibition additionally suppresses IL-18 and pyroptosis, so it is not strictly refuted — but the precedent is discouraging.
4. **Heterogeneity and stage.** Inflammasome activity is non-uniform (male/high-IL-18 enriched) and immune states flip between protective and toxic with stage/genotype (P 42638120). An all-comers trial risks diluting any real effect.

Net interpretation (INFERENCE): This is a *target-validated, drug-ready, but efficacy-unproven* repurposing hypothesis. The correct scientific question is no longer "is the mechanism real?" (it is) but "in whom, at what stage, and with what redundancy-accounting does blocking it change outcomes?"

Evidence Base

PMID	Study (abbreviated)	Role	How it supports / challenges the hypothesis
26200799	NLRP3 in SOD1 mouse & human ALS	**Supports + qualifies**	Human tissue ↑ NLRP3/ASC/IL18/ caspase-1; but astrocytes, not microglia, are the major NLRP3 cell
35867112	Pyroptosis in ALS motor cortex	**Supports**	Independent human tissue: microglial pyroptosis linked to neuronal loss (n=25 vs 19)
20616033	Mutant SOD1 IL-1β accelerates ALS	**Supports (causal)**	Caspase-1/IL-1β deletion or IL-1RA extends SOD1 mouse lifespan; strongest causal preclinical evidence
31596526	Microglial NLRP3 activated by ALS proteins	**Supports**	Establishes NLRP3 specificity linking aggregates to caspase-1/IL-1β
26222336	TDP-43 activates microglia via NF-κB & NLRP3	**Supports + challenges**	Confirms TDP-43→NLRP3/caspase-1/ IL-1β; reveals parallel NF-κB/AP-1 arms (redundancy)
26444282	Anakinra pilot in ALS	**Challenges (key)**	IL-1 blockade did not slow progression; 94% anti-drug antibodies
33257576	OLT1177 rescues cognition in AD mice	**Supports (drug)**	Oral, NLRP3-specific, human-safe, brain-penetrant, CNS-efficacious
33005902	Dapansutrile phase 2a gout	**Supports (drug)**	Oral target engagement, efficacy, tolerability in IL-1β-driven disease
33235030	Dapansutrile phase 1b heart failure	**Supports (safety)**	No serious adverse events in chronic-disease population
36774721	Masitinib in ALS	**Supports (concept)**	Neuroinflammation-targeting therapy improved ALS survival/function (different axis)
29046475	Mast cells in ALS model	**Supports (concept)**	Targeting a neuroinflammatory mechanism reduced NMJ denervation 35%
42425169		**Supports (stratifier)**	CSF IL-18 elevated, concentrated in male patients — candidate biomarker

PMID	Study (abbreviated)	Role	How it supports / challenges the hypothesis
	Sex-associated neuroinflammation in ALS		
42646494	NLRP3 in ALS PBMCs	**Supports**	Peripheral-blood NLRP3 mRNA ↑ in living patients
41832208	Canine degenerative myelopathy	**Supports**	Spontaneous ALS model: peripheral IL-18/NLRP3 ↑
42621714	Electroacupuncture NLRP3 in SOD1 mice	**Supports**	NLRP3 inhibition associated with ↑survival/function
42864293	Electroacupuncture NLRP3/Caspase-1/ASC	**Supports**	Independent corroboration of NLRP3-axis benefit
42638120	Precision medicine in ALS neuroinflammation	**Challenges (strategy)**	Immune states stage/genotype-dependent; broad immunosuppression fails
42771346	DNL343 HEALEY trial (NCT05842941)	**Cautionary context**	Mechanistically sound, target-engaged ALS drug still failed to slow progression
21082322	Respiratory management in ALS	**Safety context**	Respiratory infection is commonest cause of death — immunosuppression risk

Trial landscape (direct answer to Question 2)

A reading of the available evidence found **no registered clinical trial of dapansutrile or any other NLRP3-specific inhibitor in ALS.** The nearest human tests are:

- **Anakinra** (IL-1 receptor antagonist, downstream of NLRP3) — **negative** in a 17-patient, 1-year pilot (PMID: 26444282).
- **Masitinib** (a different neuroinflammatory axis — mast cell/microglial tyrosine kinase) — **positive** on survival/function but not NLRP3-directed (PMID: 36774721).

Dapansutrile's trial record in *related* diseases establishes safety and peripheral target engagement but **not** CNS disease modification: gout flares (phase 2a, PMID: 33005902) and NYHA II–III systolic heart failure (phase 1b, PMID: 33235030). **(INFERENCE)** These are IL-1β-driven peripheral inflammatory conditions; transfer to a chronic CNS neurodegeneration is plausible given brain penetrance (P 33257576) but not demonstrated. For broader context on the difficulty of translating mechanistically sound ALS targets, the recent DNL343 integrated-stress-response trial (NCT05842941) is instructive: despite strong proof-of-mechanism, *"200-mg/d DNL343 did not show evidence of slowing disease progression in ALS, highlighting the need for alternative therapeutic approaches"* (PMID: 42771346) — a cautionary precedent that mechanism plus target engagement does not guarantee efficacy.

Limitations and Knowledge Gaps

1. **No ALS-specific NLRP3 inhibitor data of any kind.** Every efficacy claim for dapansutrile in neurodegeneration rests on an Alzheimer's model (P 33257576), not ALS. There is zero direct in-ALS evidence that dapansutrile changes disease features — the most important gap.
2. **The one relevant human test was negative.** Anakinra (P 26444282) is the closest analog and did not slow ALS, though it was underpowered (n=17), used historical controls, acted downstream of NLRP3, and was confounded by 94% anti-drug antibodies. It constrains optimism without refuting upstream NLRP3 blockade.
3. **Cell-compartment uncertainty.** The hypothesis centers microglia, but human spinal cord data point to astrocytes as the major NLRP3-expressing cell (P 26200799).

4. **Redundancy.** TDP-43 engages NF-κB and AP-1 in parallel (P 26222336); a single-axis NLRP3 inhibitor may be insufficient.
5. **Heterogeneity and staging.** Inflammasome activity is non-uniform (male/high-IL-18 enriched; P 42425169) and immune states are stage/genotype-dependent (P 42638120). The optimal patient subgroup and treatment window are undefined.
6. **Biomarker validation is incomplete.** CSF IL-18 is a plausible stratifier but has not been prospectively validated as a predictor of NLRP3-inhibitor response in any disease, let alone ALS.
7. **Causal preclinical evidence is weighted toward SOD1.** SOD1 models dominate (PMIDs 20616033, 42621714, 42864293), yet most human ALS is TDP-43-driven and sporadic; transfer across genotypes is assumed, not demonstrated for a pharmacological NLRP3 inhibitor.
8. **Chronic safety in a respiratory-vulnerable population is unquantified.** Short-term tolerability is reassuring (PMIDs 33235030, 33005902, 26444282), but long-term infection risk in patients prone to aspiration pneumonia (P 21082322) has not been studied.

Proposed Follow-up Experiments / Actions

Priority 1 — Biomarker-stratified phase 2 trial with a target-engagement readout

Design a randomized, placebo-controlled trial of oral dapansutrile enrolling **only patients with demonstrable inflammasome activation** (e.g., CSF IL-18 above a pre-specified threshold;

).

P 42425169

Co-primary readouts: (a) **pharmacodynamic target engagement** — reduction in CSF IL-18/IL-1β from baseline, to prove the drug is hitting the axis in the CNS (the step anakinra arguably never confirmed); and (b) ALSFRS-R slope plus survival via a shared-parameter model. This directly addresses the historical failure mode of unselected anti-inflammatory ALS trials (

P 42638120

) and the DNL343 lesson (

P 42771346

) that engagement must be measured, not assumed. **This is the single most efficient experiment to test the hypothesis.**

Priority 2 — Direct preclinical test of dapansutrile in ALS models

Administer oral dapansutrile to SOD1-G93A mice (the model with existing causal NLRP3 evidence,

)

P 20616033

and a TDP-43 model, with CNS pharmacokinetics, CSF IL-18/IL-1β, survival, and motor endpoints. This fills the single largest gap — there is currently no dapansutrile-in-ALS efficacy data at all — and tests transfer across the SOD1/TDP-43 divide.

Priority 3 — Resolve the cellular compartment

Use single-cell/single-nucleus transcriptomics and immunohistochemistry on ALS spinal cord/motor cortex to quantify microglial vs astrocytic NLRP3 activity (reconciling

P 26200799

with

),

P 35867112

and confirm dapansutrile suppresses the axis in whichever cell dominates.

Priority 4 — Combination/redundancy study

Test whether NLRP3 inhibition alone is inferior to NLRP3 + NF-κB (or + cGAS-STING) blockade in TDP-43-driven microglial models (motivated by P 26222336), to determine whether single-axis therapy is doomed by redundancy before committing to monotherapy trials.

Priority 5 — Prospective validation of CSF IL-18 as a predictive biomarker

In a longitudinal ALS cohort, test whether baseline CSF IL-18 predicts progression rate and (in treated patients) response, formally qualifying the stratifier before it anchors a registrational trial.

Verdict

How strong is the case? The case for the *mechanism* is strong and multi-layered — human tissue, living-patient CSF and blood, two proteinopathy triggers, and causal mouse genetics all converge on an active NLRP3 → caspase-1 → IL-1β/IL-18 axis in ALS, and the specific drug (dapansutrile) is oral, NLRP3-selective, brain-penetrant and human-safe. The case for *efficacy in ALS* is weak: no NLRP3 inhibitor has been trialed, the nearest human test (anakinra) failed, redundant NF-κB/AP-1 arms persist, the dominant NLRP3 cell may be astrocytic rather than microglial, and immune states are stage- and genotype-dependent in a disease where broad immunosuppression has repeatedly failed. The most efficient test is a **biomarker-stratified phase 2 trial** of oral dapansutrile restricted to high-CSF-IL-18 patients, with a CSF IL-18/IL-1β target-engagement co-primary endpoint to confirm CNS pathway suppression before adjudicating ALSFRS-R and survival — rather than an all-comers trial that would likely repeat history.

Highest-weight identifiers:

- **PMID: 20616033** — Caspase-1/IL-1β deletion or IL-1RA extends SOD1 mouse lifespan: the strongest causal preclinical support.
- **PMID: 26444282** — Anakinra pilot: the only human IL-1-axis ALS trial, and it was negative.
- **PMID: 26200799** — Human ALS tissue has ↑NLRP3/ASC/IL18/caspase-1, but astrocytes dominate expression.

- **PMID: 26222336** — TDP-43 activates microglial NLRP3 but also NF-κB/AP-1 (redundancy the drug won't block).
- **PMID: 33257576** — Dapansutrile is oral, NLRP3-specific, human-safe, brain-penetrant, CNS-efficacious in AD.
- **PMID: 42425169** — CSF IL-18 elevated and male-enriched: the candidate stratifying biomarker.
- **PMID: 42638120** — Stage/genotype-dependent immune states; broad immunosuppression fails — mandates stratification.
- **PMID: 42771346** / **NCT05842941** — DNL343: mechanism + target engagement still failed, underscoring that efficacy is not guaranteed.

Separation of evidence: Findings F001–F010 and all quoted claims are ESTABLISHED from the cited primary sources. Statements explicitly labeled (INFERENCE) are my own reasoning built on those findings, not direct claims from any single abstract.
